The code

Schema SQL

```
CREATE DATABASE airport_management;
USE airport_management;

CREATE TABLE Plane_Model (
    model_no INT PRIMARY KEY,
    model_name VARCHAR(100),
    manufacturer VARCHAR(100),
    capacity INT CHECK (capacity > 0)
);

CREATE TABLE Airplane (
    plane_no INT PRIMARY KEY,
    model_no INT,
    manufacture_year INT,
    status VARCHAR(50),
    hangar_id INT,

    FOREIGN KEY (model_no)
    REFERENCES Plane_Model(model_no)
);

CREATE TABLE Hangar (
    hangar_id INT PRIMARY KEY,
    hangar_name VARCHAR(100),
    location VARCHAR(100),
    capacity INT CHECK (capacity > 0)
);

ALTER TABLE Airplane
```

```sql
ADD CONSTRAINT fk_hangar
FOREIGN KEY (hangar_id)
REFERENCES Hangar(hangar_id);

CREATE TABLE Union_Table (
    union_id INT PRIMARY KEY,
    union_name VARCHAR(100),
    membership_fee DECIMAL(10,2)
);

CREATE TABLE Employee (
    employee_id INT PRIMARY KEY,
    first_name VARCHAR(100),
    last_name VARCHAR(100),
    salary DECIMAL(10,2),
    phone VARCHAR(20),
    union_id INT,

    FOREIGN KEY (union_id)
    REFERENCES Union_Table(union_id)
);

CREATE TABLE Technician (
    technician_id INT PRIMARY KEY,
    employee_id INT,
    specialization VARCHAR(100),
    experience_years INT,

    FOREIGN KEY (employee_id)
    REFERENCES Employee(employee_id)
);
```

```sql
CREATE TABLE Traffic_Controller (
    controller_id INT PRIMARY KEY,
    employee_id INT,
    medical_exam_date DATE,
    license_number VARCHAR(50) UNIQUE,

    FOREIGN KEY (employee_id)
    REFERENCES Employee(employee_id)
);

CREATE TABLE Test (
    test_id INT PRIMARY KEY,
    test_name VARCHAR(100),
    max_score INT CHECK (max_score <= 100)
);

CREATE TABLE Testing_Event (
    event_id INT PRIMARY KEY,
    plane_no INT,
    technician_id INT,
    test_id INT,
    test_date DATE,
    score INT,
    result VARCHAR(50),

    FOREIGN KEY (plane_no)
    REFERENCES Airplane(plane_no),

    FOREIGN KEY (technician_id)
    REFERENCES Technician(technician_id),

    FOREIGN KEY (test_id)
```

```sql
    REFERENCES Test(test_id)
);

CREATE TABLE Technician_Expertise (
    expertise_id INT PRIMARY KEY,
    technician_id INT,
    model_no INT,

    FOREIGN KEY (technician_id)
    REFERENCES Technician(technician_id),

    FOREIGN KEY (model_no)
    REFERENCES Plane_Model(model_no)
);

INSERT INTO Plane_Model
VALUES
(1, 'Boeing 737', 'Boeing', 180),
(2, 'Airbus A320', 'Airbus', 170),
(3, 'Boeing 777', 'Boeing', 396);

INSERT INTO Union_Table
VALUES
(1, 'Aviation Workers Union', 250.00),
(2, 'Airport Staff Association', 180.00);

INSERT INTO Employee
VALUES
(1, 'John', 'Smith', 5000.00, '555-1111', 1),
(2, 'Emma', 'Johnson', 6200.00, '555-2222', 1),
(3, 'Michael', 'Brown', 4800.00, '555-3333', 2);
```

```sql
INSERT INTO Technician
VALUES
(1, 1, 'Engine Systems', 5),
(2, 2, 'Electrical Systems', 7);

INSERT INTO Traffic_Controller
VALUES
(1, 3, '2026-01-15', 'LIC1001');

INSERT INTO Hangar
VALUES
(1, 'North Hangar', 'Terminal A', 10),
(2, 'South Hangar', 'Terminal B', 15),
(3, 'VIP Hangar', 'Terminal C', 5);

INSERT INTO Airplane
VALUES
(1001, 1, 2020, 'Active', 1),
(1002, 2, 2021, 'Maintenance', 2),
(1003, 3, 2019, 'Active', 3);

INSERT INTO Test
VALUES
(1, 'Engine Safety Test', 100),
(2, 'Electrical Inspection', 100),
(3, 'Fuel System Test', 100);

INSERT INTO Testing_Event
VALUES
(1, 1001, 1, 1, '2026-05-10', 95, 'Passed'),
(2, 1002, 2, 2, '2026-05-11', 88, 'Passed'),
(3, 1003, 1, 3, '2026-05-12', 76, 'Needs Repair');
```

```
INSERT INTO Technician_Expertise
VALUES
(1, 1, 1),
(2, 1, 3),
(3, 2, 2);
```

Query SQL

```
SELECT * FROM Airplane;
SELECT * FROM Employee;
SELECT * FROM Testing_Event;
SELECT first_name, last_name, specialization
FROM Employee
JOIN Technician
ON Employee.employee_id = Technician.employee_id;
SELECT plane_no, model_name, manufacturer
FROM Airplane
JOIN Plane_Model
ON Airplane.model_no = Plane_Model.model_no;

SELECT COUNT(*) AS total_airplanes
FROM Airplane;

SELECT AVG(score) AS average_score
FROM Testing_Event;

SELECT MAX(capacity) AS largest_capacity
FROM Plane_Model;

SELECT manufacturer, COUNT(*) AS total_models
FROM Plane_Model
```

```sql
GROUP BY manufacturer;

SELECT result, COUNT(*) AS total_results
FROM Testing_Event
GROUP BY result;

SELECT first_name, last_name, salary
FROM Employee
ORDER BY salary DESC;

SELECT specialization, COUNT(*) AS total_technicians
FROM Technician
GROUP BY specialization;

SELECT model_name, capacity
FROM Plane_Model
WHERE capacity > 200;

SELECT plane_no, score
FROM Testing_Event
WHERE score > 80;
```